Name: _____ Date: _____ Period: _____

HONORS CHEMISTRY • UNIT 2

Electrons & Periodic Trends

The arrangement of electrons explains the shape of the periodic table, why elements react the way they do, and every trend you will use for the rest of the course.

HOW TO USE THIS PACKET

This is your notebook for the unit. Write definitions in your own words, copy the steps of each worked example, and use the blank space for sketches and questions. Boxes marked LAB CORNER are things you would do or see at a lab bench. Extra practice is at the back.

What we will learn

2.1	Bohr models & Lewis dot structures
2.2	The quantum mechanical model & electron configuration
2.3	Orbital diagrams (Aufbau, Pauli & Hund)
2.4	Periodic trends: ionization energy & atomic radius
2.5	Periodic trends: electronegativity & successive ionization energy
★	Extra practice & unit review

By the end of this unit I can...

- Explain how Bohr's model improved on Rutherford's, and use line spectra and flame tests as evidence for energy levels.
- Draw Bohr models and Lewis dot structures and find valence electrons from a periodic table.
- Write full and noble-gas shorthand electron configurations for atoms and ions.
- Draw orbital diagrams that obey the Aufbau principle, the Pauli exclusion principle and Hund's rule.
- Use Coulomb's law (nuclear charge, distance and shielding) to explain trends in ionization energy and atomic radius.
- Explain the trend in electronegativity and use successive ionization energies to identify an element's group.
- Rank atoms and ions in order of radius, ionization energy and electronegativity and justify each ranking.

Name: _____ Date: _____ Period: _____

TOPIC
2.1

Bohr Models & Lewis Dot Structures

Energy levels • line spectra • valence electrons

BIG IDEA

Bohr proposed that electrons can only exist in certain **energy levels** (shells) around the nucleus, like steps on a staircase. Electrons jump between levels by absorbing or emitting exactly the right amount of energy.

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
Energy level (shell)	*How many e^- can the first three levels hold?*	
Valence electron	*Where are they? Why do they matter?*	
Core electron	*How is it different from a valence electron?*	
Emission spectrum	*What makes the lines?*	
Lewis dot structure	*What do the dots stand for?*	

Name: _____ Date: _____ Period: _____

How did Bohr improve Rutherford's model?

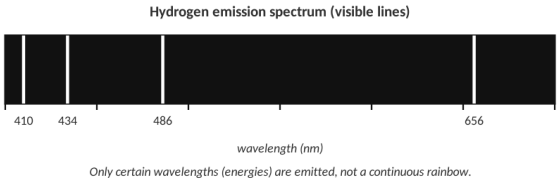
CUES / KEY WORDS	COMPARE THE TWO MODELS
Rutherford **Bohr** **Evidence**	

THINK OF IT LIKE... A STAIRCASE

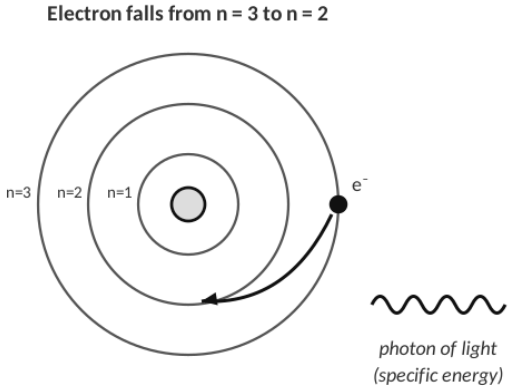
On a staircase you can stand on step 1 or step 2, but never in between. Electrons are the same: they can only be at certain energy levels. Moving up a step takes a specific amount of energy; moving down releases the same amount, as light.

LAB CORNER • Line spectra: the fingerprint of an element

When hydrogen gas is heated or energized, it glows. A prism or diffraction grating spreads that light into a spectrum.



Visible lines in the hydrogen spectrum



An electron falls to a lower level

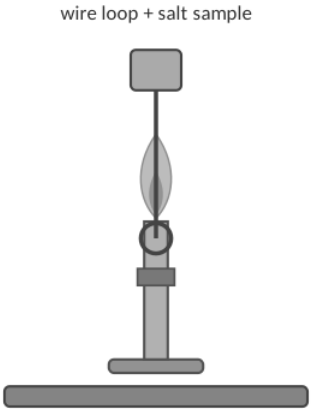
1. Why does hydrogen emit only a few separate lines and not a continuous rainbow of colors?

2. The line at 410 nm has a shorter wavelength than the line at 656 nm. Which line comes from the larger energy drop? Explain.

3. What happens to an electron when an atom absorbs energy? What happens when it falls back?

LAB CORNER • Flame tests

A student dips a clean wire loop into solutions of metal salts and holds it in a Bunsen burner flame. Each metal ion produces a different flame color.



Each metal ion colors the flame differently because each has its own set of electron energy levels.

Metal ion in the salt	Flame color observed
Lithium (Li^+)	crimson red
Sodium (Na^+)	bright yellow-orange
Potassium (K^+)	pale lilac / violet
Calcium (Ca^{2+})	orange-red
Copper (Cu^{2+})	blue-green
Barium (Ba^{2+})	yellow-green

4. Explain, using electron energy levels, where the colored light in a flame test comes from and why each metal gives a different color.

5. A student tests a white powder and sees a bright yellow-orange flame. What metal ion is most likely in the powder? Why isn't flame color alone enough to be certain when the flame is orange-red?

Drawing Bohr models

BOHR MODEL RULES

- The number in the nucleus = atomic number (protons).
- Fill the shells closest to the nucleus first: **2 → 8 → 8 ...** (up to element 20).
- The electrons in the last shell are the **valence electrons**.

Draw the Bohr model for each element. Write the number of protons in the nucleus and place dots for electrons on the rings.



lithium (Li)



nitrogen (N)



magnesium (Mg)



chlorine (Cl)



calcium (Ca)



boron (B)

6. How do you find the number of valence electrons for an element using the periodic table?

Group	1	2	13	14	15	16	17	18
Valence electrons								

Lewis dot structures

> **THINK OF IT LIKE... SEATS ON A SQUARE TABLE**
>
> Put one dot on each side of the symbol (top, right, bottom, left) before you pair any up. This is like people sitting at a square table: everyone takes their own side first.

Name: _____ Date: _____ Period: _____

Draw the Lewis dot structure for each element.

Li	Be	N	F	Ar	Al

Identify the element

Clue	Element
Has 5 valence electrons and is in period 3.	
Has 7 valence electrons and its most abundant isotope has 18 neutrons.	
Has 8 valence electrons and is in period 2.	
Has a Bohr model with 2, 8, 2 electrons in its shells.	

WATCH OUT!

Valence electrons are counted by the **outermost shell**, not by the total number of electrons. Sodium has 11 electrons but only **1** valence electron.

[IMAGE GOES HERE]

Optional: Bohr model diagrams from your Topic 2.1 homework (identify the elements)

7. An electron in an atom absorbs energy. What happens next?

A) It moves to a lower energy level.
B) It moves to a higher energy level.
C) It leaves the nucleus.
D) It becomes a proton.

Name: _____ Date: _____ Period: _____

TOPIC
2.2

The Quantum Mechanical Model & Electron Configuration

Sublevels • orbitals • Aufbau order

BIG IDEA

Electrons don't circle the nucleus like planets. They occupy **orbitals**, 3-D regions where an electron is most likely to be found. Orbitals are grouped into **sublevels (s, p, d, f)** inside each energy level. An **electron configuration** lists where every electron is.

Key vocabulary — write each definition in your own words

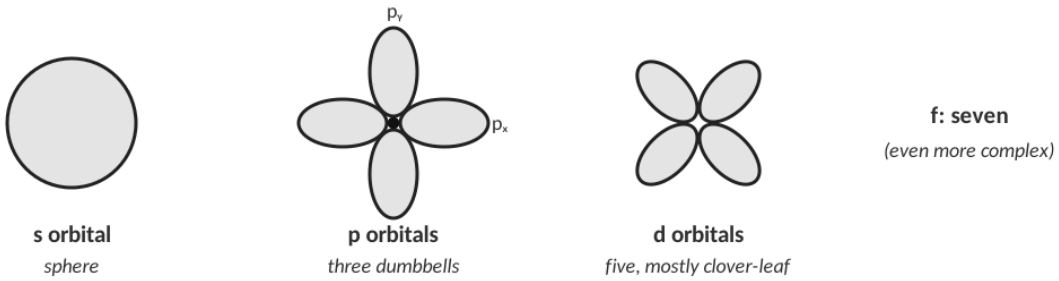
Term	Definition (your words)	Example / sketch
Orbital	*How many electrons fit in one?*	
Sublevel (s, p, d, f)	*How many orbitals in each?*	
Electron configuration	*What does $2p^4$ mean?*	
Aufbau principle	*Which sublevels fill first?*	
Noble gas shorthand	*Why do we use it?*	

THINK OF IT LIKE… AN APARTMENT BUILDING

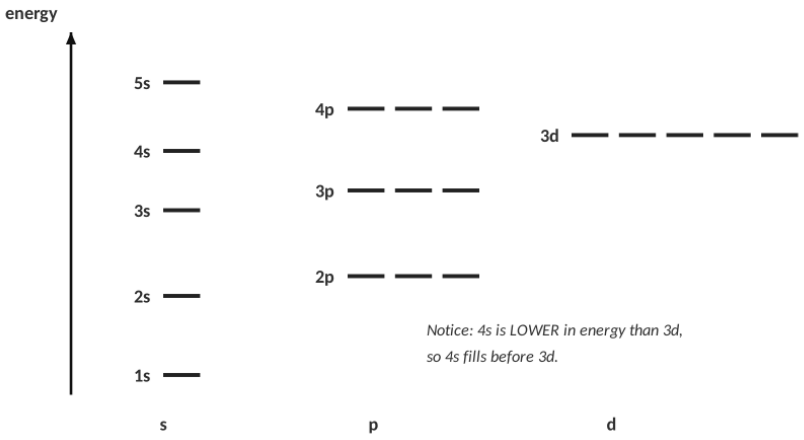
The **energy level** is the floor of the building (n = 1, 2, 3 …). Each floor has different **types of rooms** (s, p, d, f). Each room is an **orbital**, and every room holds at most 2 people. People take the cheapest, lowest rooms first, and the building gets bigger as you go up (floor 1 has just an s room; floor 3 has s, p and d rooms).

Name: _____ Date: _____ Period: _____

Sublevel	Shape	Number of orbitals	Max electrons
s			
p			
d			
f			

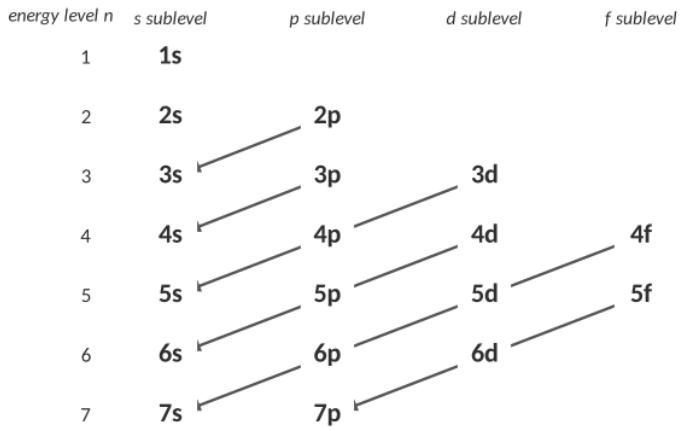


Orbital shapes (not to scale).



Relative energies of the sublevels.

Name: _____ Date: _____ Period: _____



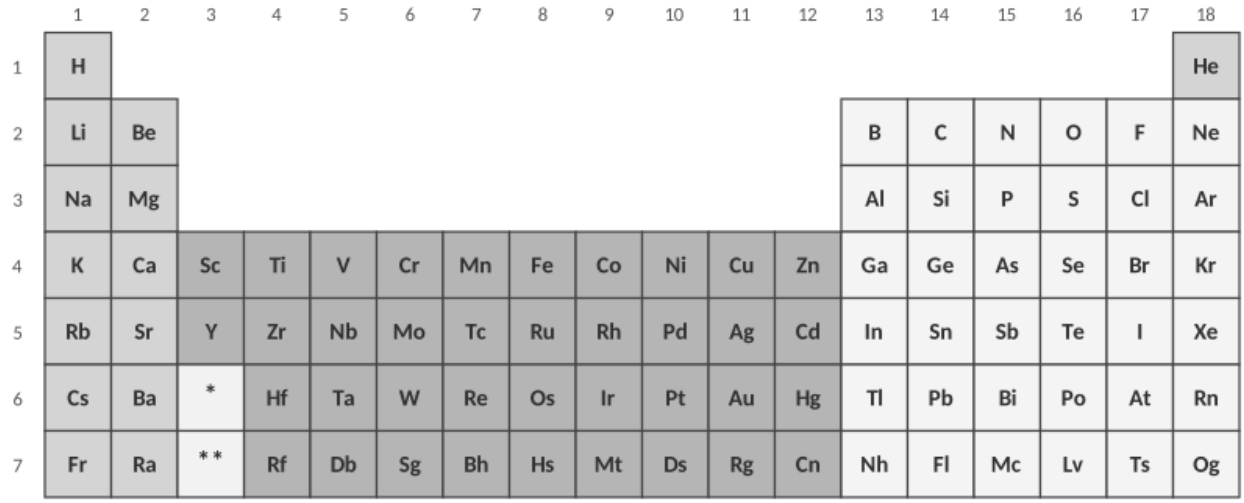
Fill in this order:

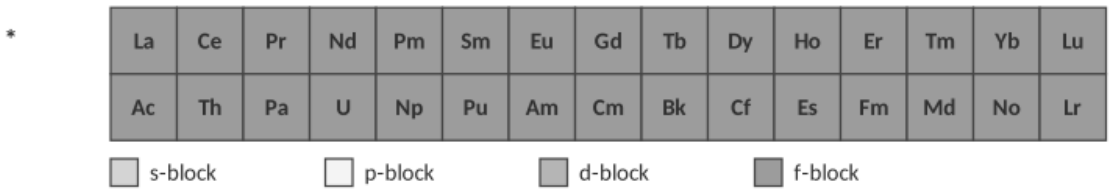
1s 2s 2p 3s 3p
4s 3d 4p 5s 4d
5p 6s 4f 5d 6p
7s 5f 6d 7p

The diagonal rule: follow each arrow from top right to bottom left.

CUES / KEY WORDS	WRITE THE FILLING ORDER FROM MEMORY (USE THE DIAGRAM AS A GUIDE)
Aufbau **Order**	

The periodic table is a map of the filling order





Blocks of the periodic table.

1. Use the table above to explain why the periodic table has a block of 2 columns on the left, 6 on the right, and 10 in the middle.

Worked examples

WORKED EXAMPLE — FULL CONFIGURATION

Write the electron configuration for nitrogen (7 electrons).

Work it out here:

Answer: _____

WORKED EXAMPLE — SHORTHAND

Write the noble gas shorthand for arsenic (As, 33 electrons).

Work it out here:

Answer: _____

WORKED EXAMPLE — IDENTIFY FROM THE CONFIGURATION

Which element has the configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$?

Work it out here:

Answer: _____

QUICK TRICKS

- The **period number** is the highest energy level (n) in the configuration.
- **Valence electrons** = electrons in the highest energy level (the s and p electrons of the outer shell).
- Superscripts always add up to the atomic number (for a neutral atom).

Name: _____ Date: _____ Period: _____

Practice: write the full configuration, then circle the valence electrons

Element	Electron configuration	Valence e^-
Be (Z = 4)		
Mg (Z = 12)		
Ca (Z = 20)		

2. What do the configurations of Be, Mg and Ca have in common? Why does this make sense?

Element	Electron configuration	Valence e^-
N (Z = 7)		
Cl (Z = 17)		
Mn (Z = 25)		
Cd (Z = 48)		

Practice: noble gas shorthand

Element	Shorthand configuration
Sr (Z = 38)	
As (Z = 33)	
S (Z = 16)	
Al (Z = 13)	

Name: _____ Date: _____ Period: _____

Practice: identify the element

Configuration	Element
$1s^2\ 2s^2\ 2p^6\ 3s^1$	
$[Kr]\ 5s^2\ 4d^{10}\ 5p^3$	
$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^6\ 5s^2\ 4d^{10}\ 5p^3$	
$[Ne]\ 3s^2\ 3p^2$	

Find the error

Each configuration below is for a neutral atom. Is it correct? If not, explain what is wrong and write the correct one.

Configuration	Correct? If not, fix it
$1s^2\ 2s^2\ 2p^6\ 3s^3$	
$1s^2\ 2s^2\ 2p^7$	
$1s^2\ 2s^1\ 2p^5$	
$[Ar]\ 4s^2\ 3d^{10}\ 4p^5$	
$[Ar]\ 5s^2\ 4d^{10}\ 5p^5$	

Configurations of ions

CUES / KEY WORDS	HOW DO YOU WRITE THE CONFIGURATION OF AN ION?
Cations **Anions** **Isoelectronic**	

Ion	Configuration	Same configuration as which noble gas?
Na^+		
O^{2-}		
Mg^{2+}		
Cl^-		
Ca^{2+}		
Al^{3+}		

REAL WORLD

Atoms "want" a filled outer shell like a noble gas. That is why sodium readily forms Na^+ and chlorine forms Cl^-: they end up with the same stable configuration as neon and argon. This is the whole story of ionic bonding (Unit 9).

3. How many electrons can the 3d sublevel hold?

A) 2
B) 6
C) 10
D) 14

4. Which is the correct shorthand configuration for calcium?

A) [Ne] $3s^2$
B) [Ar] $4s^2$
C) [Ar] $3d^2$
D) [Kr] $4s^2$

Name: _____ Date: _____ Period: _____

TOPIC
2.3

Orbital Diagrams

Aufbau • Pauli exclusion • Hund's rule

BIG IDEA

An **orbital diagram** shows every orbital as a box and every electron as an arrow. It tells you more than a configuration: it shows which electrons are **paired** and which are **unpaired**, and that decides how an atom behaves in a magnetic field and in bonding.

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
Orbital	*How many electrons can one orbital hold?*	
Pauli exclusion principle	*What must be different about two electrons in the same box?*	
Hund's rule	*What do electrons do in a sublevel with several boxes?*	
Paired / unpaired electron	*How can you tell from a diagram?*	
Degenerate orbitals	*Which orbitals are degenerate?*	

THINK OF IT LIKE… SEATS ON A BUS

Strangers boarding a bus take an empty double-seat to themselves before sitting next to someone (**Hund's rule**). Only when every double-seat has one person does anyone sit beside a stranger. Each seat holds at most two people, and they must be facing opposite directions (**Pauli**). Passengers fill the front rows first (**Aufbau**).

Name: _____ Date: _____ Period: _____

The three rules — write each one in your own words

CUES / KEY WORDS	STATE EACH RULE AND SKETCH A SMALL EXAMPLE
Aufbau principle **Pauli exclusion principle** **Hund's rule**	

How to draw an orbital diagram

STEPS

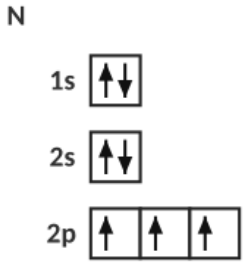
1. Write the electron configuration first (so you know how many electrons go in each sublevel).
2. Draw boxes: **1 box for s, 3 for p, 5 for d, 7 for f**.
3. Add arrows one at a time, ↑ first in every box of the sublevel (Hund), then add ↓ to pair them.
4. Check: the total number of arrows equals the atomic number, and no box holds more than two arrows.

WORKED EXAMPLE — NITROGEN

Draw the orbital diagram for nitrogen (Z = 7).

Work it out here:

Answer: _____



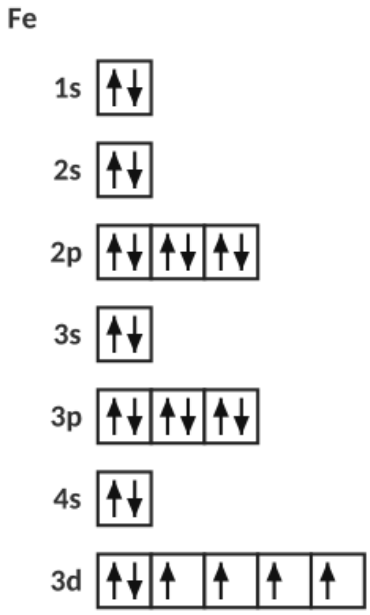
Nitrogen: three unpaired electrons.

WORKED EXAMPLE — IRON

Draw the orbital diagram for iron (Z = 26) and count its unpaired electrons.

Work it out here:

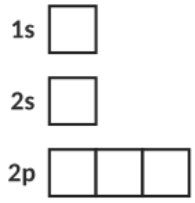
Answer: _____



Iron: four unpaired electrons in 3d.

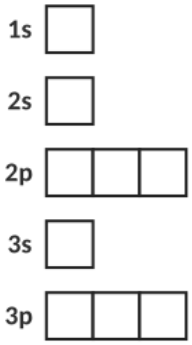
Practice: complete the orbital diagram

Fill in the arrows. Then write the number of unpaired electrons.



O (Z = 8): unpaired electrons = _____

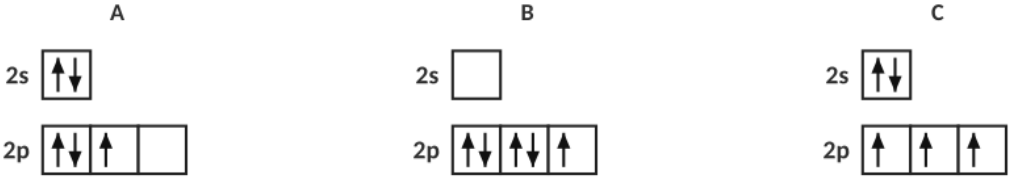
Name: _____ Date: _____ Period: _____



P (Z = 15): unpaired electrons = _____

Find the error

Each diagram below is meant to show nitrogen's outer electrons (2s and 2p). Only one is correct. For the other two, name the rule that is broken.



1. Which diagram is correct, and what rule is broken in each of the others?

Unpaired electrons across period 3

Element	Shorthand configuration	Unpaired e^-	Attracted to a magnet?
Na			
Mg			
Al			
Si			
P			
S			

Name: _____ Date: _____ Period: _____

Element	Shorthand configuration	Unpaired e^-	Attracted to a magnet?
Cl			
Ar			

2. Describe the pattern in the number of unpaired electrons as you move across period 3. Why does it go up and then back down?

Orbital diagrams for ions

3. Draw the 2p orbital boxes for O ($2p^4$) and for O^{2-}. How many unpaired electrons does each have? Why is O^{2-} stable?

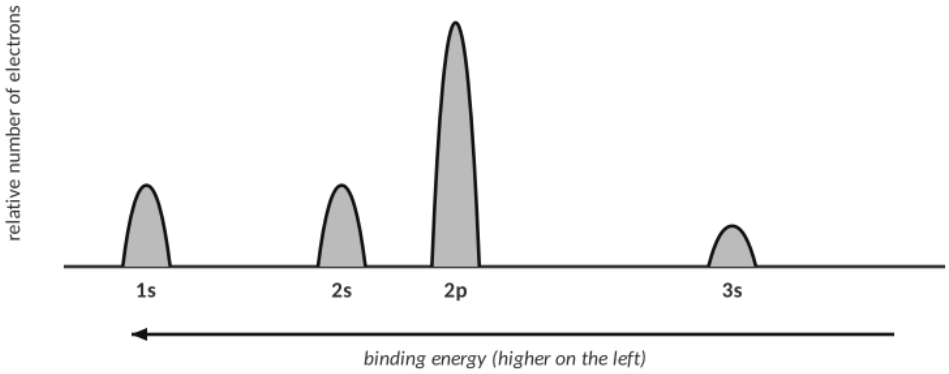
4. Write the electron configuration and count the unpaired electrons for Fe^{2+} and Fe^{3+}. (Remove the 4s electrons first.)

Extension: photoelectron spectroscopy (PES) as evidence for sublevels

In PES a sample is hit with high-energy light and the energy needed to remove each electron is measured. Each peak is one sublevel; the **height** is the number of electrons in it. Peaks farther **left** need more energy (those electrons are closest to the nucleus).

Name: _____ Date: _____ Period: _____

Photoelectron spectrum of sodium



5. The PES for an unknown element shows three peaks with relative heights 2 : 2 : 1 (left to right). Which element is it? Write its configuration.

6. In the sodium spectrum, why is the 1s peak farthest to the left?

7. How many unpaired electrons does a ground-state atom of sulfur have?

A) 0
B) 1
C) 2
D) 4

8. Which rule is broken if two electrons in the same orbital both point up?

A) Aufbau
B) Hund
C) Pauli exclusion
D) Dalton

9. Which element has exactly three unpaired electrons?

A) Si
B) P
C) S
D) Cl

Name: _____ Date: _____ Period: _____

TOPIC
2.4

Periodic Trends I: Ionization Energy & Atomic Radius

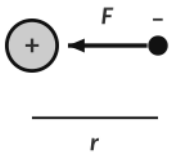
Coulomb's law • nuclear charge • distance • shielding

BIG IDEA

Almost every periodic trend comes from one idea: **opposite charges attract.** The nucleus (+) pulls on the electrons (−). How hard it pulls depends on **how many protons**, **how far away** the electron is, and **how many other electrons are in the way**.

Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
Atomic radius	*How is it measured, and in what units?*	
Ionization energy (IE)	*Does it take energy or release it?*	
Effective nuclear charge (Zeff)	*Is it larger or smaller than the real nuclear charge?*	
Shielding	*Which electrons do the shielding?*	
Coulomb's law	*What are the three factors?*	



$$F = k\,(q_1\,q_2) / r^2$$

bigger charges (q) → stronger attraction

greater distance (r) → much weaker attraction

(r is squared, so distance matters a lot)

Coulomb's law: the pull on an electron.

Name: _____ Date: _____ Period: _____

THINK OF IT LIKE… A MAGNET AND A PAPERCLIP

A paperclip held close to a magnet jumps to it. Move it a little farther away and the pull almost disappears (distance). Use a stronger magnet and the pull grows (nuclear charge). Put a thick book between them and the pull is weakened (shielding). An outer electron feels the nucleus exactly like that paperclip.

The three factors — your own words

CUES / KEY WORDS	HOW DOES EACH FACTOR CHANGE THE PULL ON AN OUTER ELECTRON?
Nuclear charge **Distance** **Shielding**	

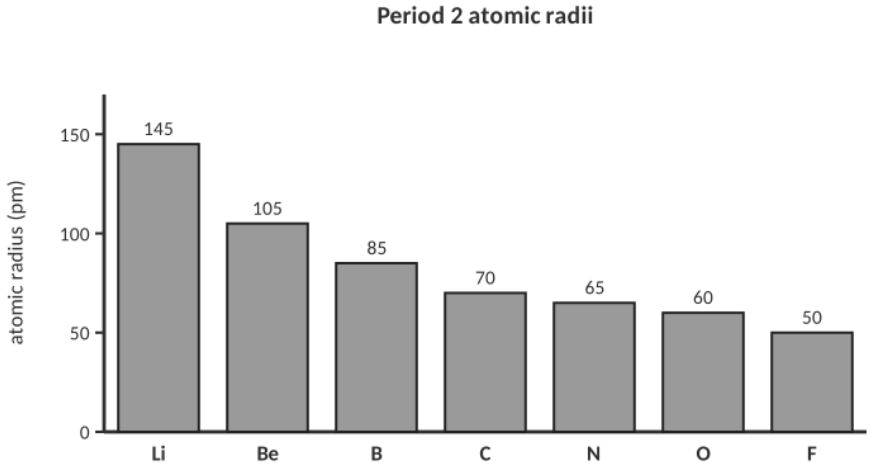
Effective nuclear charge across period 3

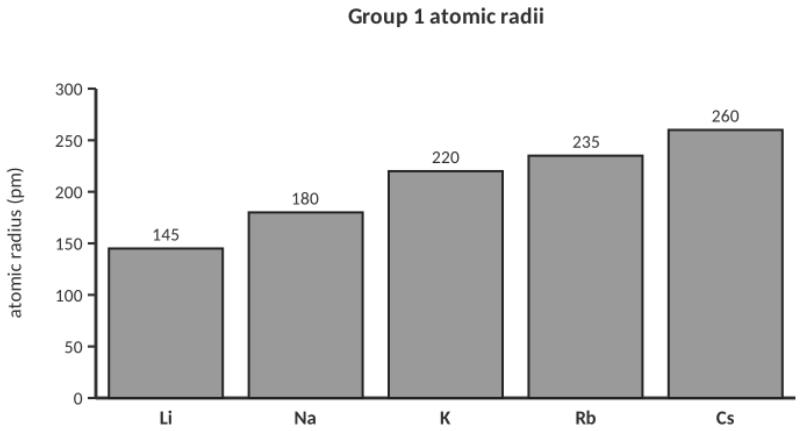
Element	Protons	Core (inner) e^-	Zeff ≈ protons − core	Valence e^-
Na	11	10		
Mg	12	10		
Al	13	10		
Si	14	10		
P	15	10		
S	16	10		
Cl	17	10		
Ar	18	10		

1. What happens to the effective nuclear charge as you go left to right across a period? Does the number of energy levels change?

Name: _____ Date: _____ Period: _____

Atomic radius





Radii shown are empirical atomic radii (pm); different sources give slightly different numbers but the same trends.

2. State the trend in atomic radius (a) across a period and (b) down a group.

Write a full explanation (CER)

CER FRAME

Claim: what is the trend or the answer?

Evidence: quote a fact, a number or a position on the table.

Reasoning: use nuclear charge, distance and shielding (Coulomb's law) to explain WHY.

3. Explain why a sodium atom is larger than a chlorine atom, using principles of atomic structure and Coulomb's law.

4. Explain why a potassium atom is larger than a sodium atom, using Coulomb's law.

Name: _____ Date: _____ Period: _____

Bohr models: lithium and sodium

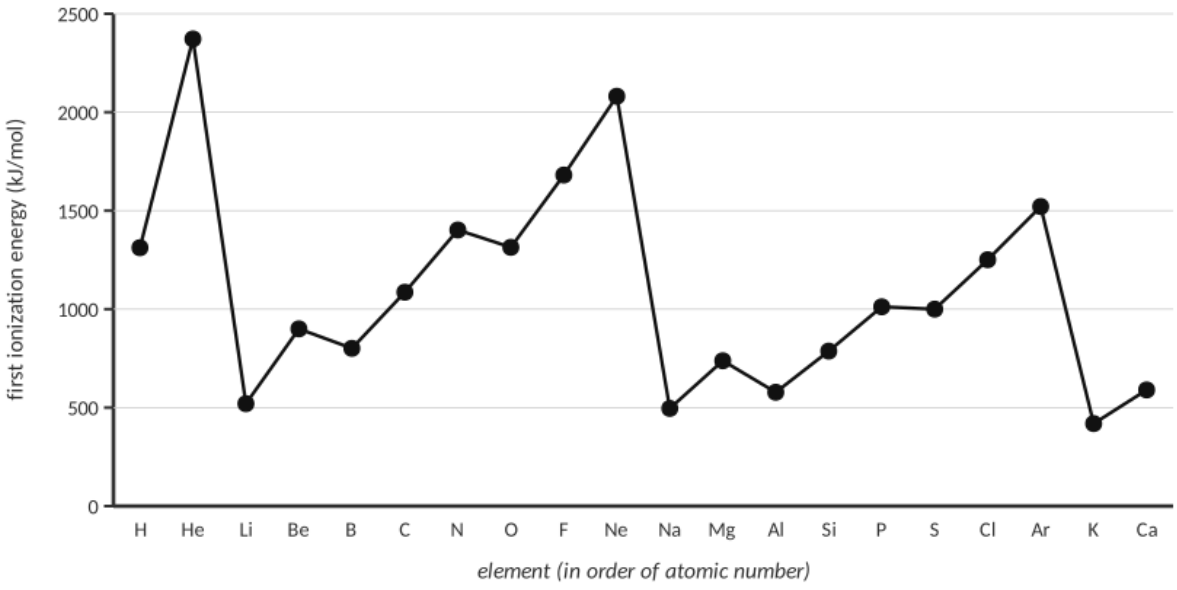
Draw Bohr models for Li and Na. Show the correct number of electrons in each shell and circle the valence electron.



Li Na

5. Using your models, explain which atom holds its valence electron more tightly.

First ionization energy



First ionization energy of elements 1–20.

6. Describe the overall pattern across a period and down a group in the graph.

7. Explain why the ionization energy drops sharply from neon to sodium.

8. Explain why helium has the largest first ionization energy of all the elements.

Two small exceptions

Look at the graph again: boron is slightly lower than beryllium, and oxygen is slightly lower than nitrogen. Use the configurations to explain one of them.

9. Explain why oxygen's first IE is lower than nitrogen's.

Name: _____ Date: _____ Period: _____

Ranking practice

Rank from lowest to highest	Answer
First ionization energy: Li, Be, B, N, O, F	
Atomic radius (smallest to largest): N, P, As, Sb, Bi, Mc	
Atomic radius (smallest to largest): Na, Mg, Al, Si, Cl	
First ionization energy (lowest to highest): Cs, K, Na, Li	

10. For each ranking above, circle the factor that matters most: nuclear charge, distance or shielding. Justify the second one in a sentence.

11. Which atom has the largest atomic radius?

A) Mg
B) Cl
C) Al
D) Na

12. Which of these has the lowest first ionization energy?

A) Li
B) Na
C) K
D) Be

13. The main reason ionization energy increases left to right across a period is that…

A) there are more energy levels
B) effective nuclear charge increases
C) shielding increases a lot
D) atoms get larger

Name: _____ Date: _____ Period: _____

TOPIC

2.5

Periodic Trends II: Electronegativity & Successive Ionization Energy

Pull in a bond • ions • families

BIG IDEA

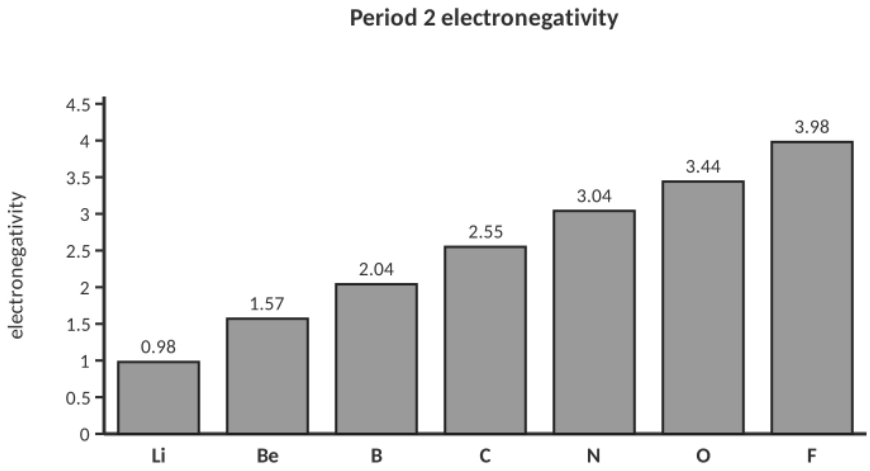
Electronegativity tells you how strongly an atom pulls on shared electrons in a bond. **Successive ionization energies** tell you how many valence electrons an atom has: the jump in energy shows where the core begins.

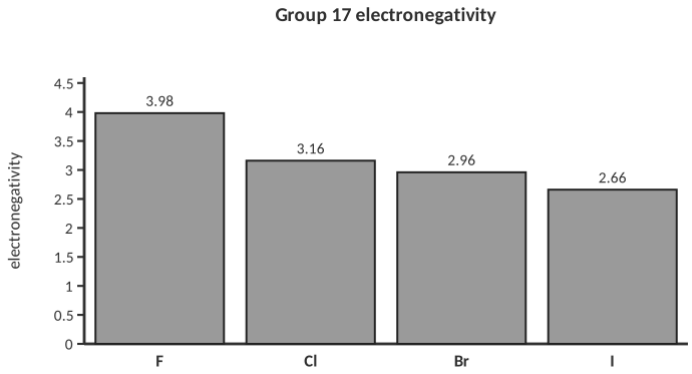
Key vocabulary — write each definition in your own words

Term	Definition (your words)	Example / sketch
Electronegativity (EN)	*In which situation is it used?*	
Successive ionization energies	*What do IE_1, IE_2, IE_3 mean?*	
Cation	*What happens to its size?*	
Anion	*What happens to its size?*	
Isoelectronic	*What do the species share?*	

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Electronegativity





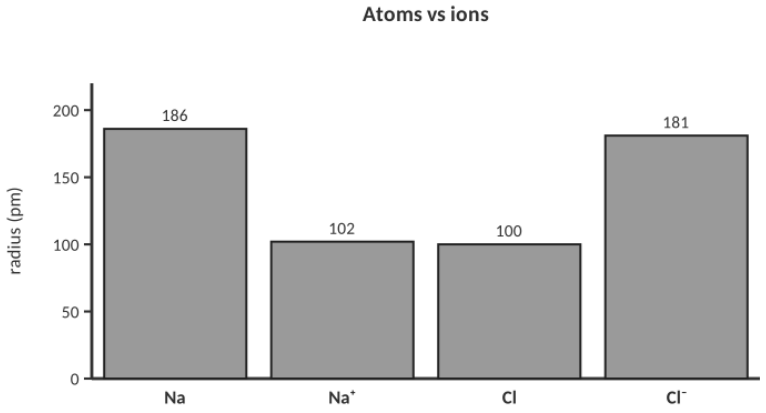
1. State the trend in electronegativity across a period and down a group. Which element is the most electronegative?

2. Explain the trend across a period using nuclear charge, distance and shielding.

3. Why don't the noble gases usually have electronegativity values?

Rank (lowest to highest EN)	Answer
Na, Cl, Al, P	
O, S, Se, Te	
C, N, O, F	

Ionic radius: why are cations small and anions big?



4. Explain why Na^+ is smaller than Na and Cl^- is larger than Cl.

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Isoelectronic series

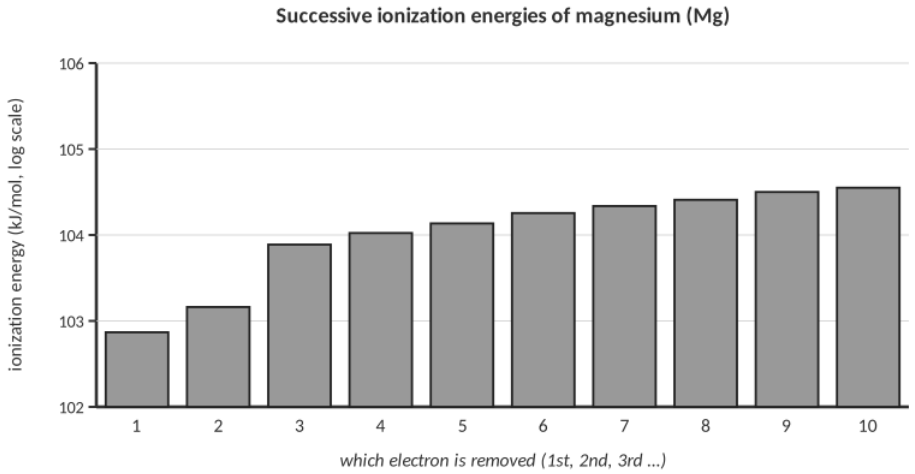
Ion	Protons	Electrons	Configuration
N^{3-}			
O^{2-}			
F^{-}			
Na^{+}			
Mg^{2+}			
Al^{3+}			

5. Rank these six ions from largest to smallest radius and explain the pattern.

Successive ionization energies

> **THINK OF IT LIKE... PEELING LAYERS**
>
> Imagine unwrapping a gift in several layers. The outside layers come off easily, but once you reach the core of the gift it is held tightly. For an atom, the valence electrons are the easy outer layers; the core electrons are very hard to remove, so you see a huge jump in energy.

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6. Where is the big jump in the Mg data, and what does it tell you about magnesium?

An unknown element's successive ionization energies (kJ/mol) are:

IE_1	IE_2	IE_3	IE_4	IE_5
786	1577	3232	4356	16091

7. In which group does this element belong? Explain, and name one possible element.

IE_1	IE_2	IE_3	IE_4	IE_5
590	1145	4912	6491	8153

8. Repeat for this second unknown.

Fluorine versus iodine

9. The first ionization energy of fluorine is 1681 kJ/mol and iodine is 1008 kJ/mol. Explain why fluorine's is so much higher.

Why do members of a family act alike?

10. Explain, using electron configurations, why sodium and potassium react in similar ways with water.

Word bank review

Fill in the blanks using the word bank: decreases, increases, nuclear charge, shielding, distance, valence, core.

11. As you move left to right across a period, atomic radius _____ because the _____ increases while the number of energy levels stays the same.

12. As you move down a group, ionization energy _____ because the valence electrons are farther away and more _____ by core electrons.

13. A big jump in successive ionization energies shows you have started removing a _____ electron.

14. Electronegativity _____ from left to right across a period.

15. Which species has the smallest radius?

A) Na^{+}
B) Mg^{2+}
C) F^{-}
D) O^{2-}

16. An element has IE_1 through IE_6 of 578, 1817, 2745, 11577, 14842, 18379. It is probably in group…

A) 1
B) 2
C) 13
D) 14

17. Which atom is the most electronegative?

A) O
B) N
C) Cl
D) S

Name: _____ Date: _____ Period: _____

PRACTICE

Extra Practice — Unit 2

Work these on your own or with a partner. Show all work.

A. Electron configurations

Element	Full configuration	Noble-gas shorthand	Valence e^-
Li (Z = 3)			
F (Z = 9)			
Si (Z = 14)			
K (Z = 19)			
Ti (Z = 22)			
Ga (Z = 31)			
Br (Z = 35)			
I (Z = 53)			

Ions

Ion	Configuration	Isoelectronic with
K^+		
S^{2-}		
Br^-		
Sr^{2+}		
Li^+		
P^{3-}		

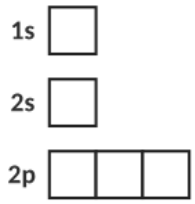
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Identify the element

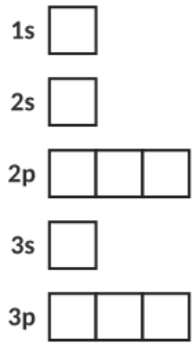
Clue	Element
$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$	
$[Ar]\ 4s^2\ 3d^{10}\ 4p^2$	
$[Kr]\ 5s^2$	
Period 3 element with 6 valence electrons	
Halogen with the configuration ending $4p^5$	
The alkali metal with four energy levels	

B. Orbital diagrams

Draw the orbital diagram for each atom and write the number of unpaired electrons.

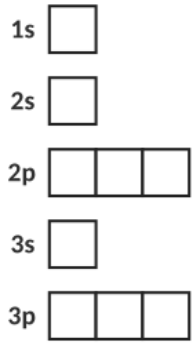


C (Z = 6): unpaired electrons = _____



S (Z = 16): unpaired electrons = _____

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Cl (Z = 17): unpaired electrons = _____

1. Atoms of which element in period 2 have the most unpaired electrons? How many?

2. Explain, using Hund's rule, why carbon's two 2p electrons are in different orbitals.

C. Bohr models and Lewis dot structures

Atom	Protons	Electrons per shell	Valence e^-	Lewis dot structure (draw)
Mg				
O				
Ar				
B				
Ca				

3. Why do elements in the same group have the same number of valence electrons? Give an example.

D. Trends

Rank	Answer
Atomic radius, smallest to largest: S, Na, Cl, Mg	
Atomic radius, smallest to largest: Ca, Be, Mg, Sr	
First ionization energy, lowest to highest: Ar, Na, Mg, Cl	
First ionization energy, lowest to highest: Ne, Kr, He, Ar	
Electronegativity, lowest to highest: P, Cl, Si, Na	
Ion radius, smallest to largest: S^{2-}, Ca^{2+}, K^{+}, Cl^{-}	

4. For the last ranking, explain in a sentence or two why it is the order you chose.

5. Explain why magnesium has a higher first ionization energy than sodium. Use Coulomb's law.

6. Explain why the atomic radius of chlorine is smaller than that of sulfur.

7. Explain why the first ionization energy of lithium is higher than that of cesium.

E. Data analysis

The table lists the first four ionization energies (kJ/mol) of four unknown elements, W, X, Y and Z, from period 3.

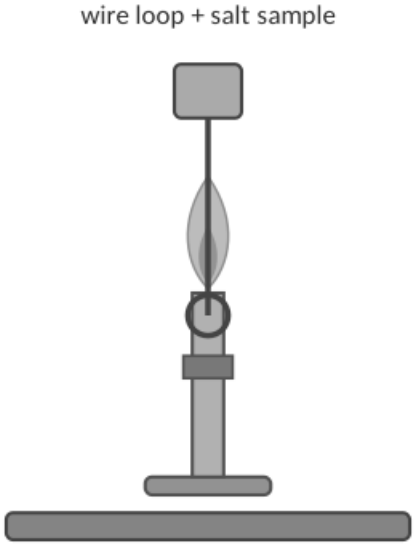
Element	IE_1	IE_2	IE_3	IE_4	Group?
W	496	4562	6910	9543	
X	738	1451	7733	10543	
Y	578	1817	2745	11577	
Z	786	1577	3232	4356	

8. Complete the last column, then name each element (all are in period 3).

9. Which element would form a +3 ion most easily? Explain using the data.

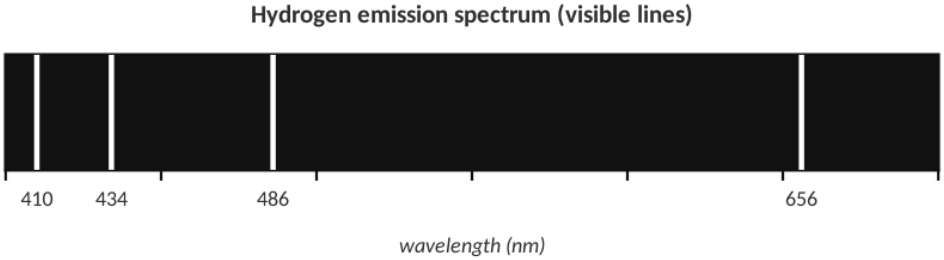
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F. Lab corner: flame tests and line spectra



Each metal ion colors the flame differently because each has its own set of electron energy levels.

Flame test: the same metal ions are held in a flame on a wire loop; the flame colour identifies the metal.



Only certain wavelengths (energies) are emitted, not a continuous rainbow.

Emission spectrum of hydrogen (bright lines on a dark background).

10. A lab student puts a salt in a flame and sees a red-orange glow. A second salt gives green. Explain at the level of electrons what is happening to produce the light.

11. Why does hydrogen give lines rather than a continuous rainbow?

12. The student wants to identify an unknown metal. Explain why a flame test or a line spectrum is better evidence than colour of the solid.

[IMAGE GOES HERE]

Add AP or homework figure: periodic table or Bohr diagram

G. Quick checks (multiple choice)

13. Which electron configuration represents an excited state of neon?

A) $1s^2\ 2s^2\ 2p^6$
B) $1s^2\ 2s^2\ 2p^5\ 3s^1$
C) $1s^2\ 2s^2\ 2p^6\ 3s^1$
D) $1s^2\ 2s^2\ 2p^4$

14. Which of the following has the same electron configuration as an Ar atom?

A) Na^+
B) Cl^-
C) F^-
D) Mg^{2+}

15. The ion with the largest radius is

A) Mg^{2+}
B) Na^+
C) F^-
D) O^{2-}

16. Which sublevel is filled right after 4s?

A) 4p
B) 3d
C) 4d
D) 5s

17. Which element has the highest first ionization energy?

A) Na
B) Mg
C) Al
D) Cl

Name: _____ Date: _____ Period: _____

TOPIC

★

Unit 2 Review

Put it in your own words before the test

1. In three or more sentences, explain how Bohr's model improved on Rutherford's and what evidence supports energy levels.

2. What do the three rules (Aufbau, Pauli, Hund) say, and how do they work together to build an orbital diagram?

3. Describe the three factors in Coulomb's law that explain periodic trends and tell how each one changes across a period and down a group.

4. Explain how successive ionization energy data identify an element's group.

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Draw a periodic table sketch and label the trends

Sketch: